

CONCEPT QUESTIONS – CH : 1

Thermodynamics & Heat Transfer

1-1C

How does the science of heat transfer differ from the science of thermodynamics?

Thermodynamics is the amount of energy transferred between equilibriums.

Heat transfer is the rate of heat transfer and the resulting temperature distributions within a system as a function of time and position.

1-2C

What is the driving force for (a) heat transfer, (b) electric current flow, and (c) fluid flow?

(a) The driving force for heat transfer is temperature difference.

(b) The driving force for electric current flow is electrical potential difference (voltage difference)

(c) The driving force for fluid flow is pressure difference

1-3C

What is the caloric theory? When and why was it abandoned?

Treated heat as a fluid substance that flowed from high temperature to low temperature. The theory stated that caloric was conserved and could neither be created nor destroyed.

The caloric theory was abandoned in the mid 19th century after experiments showed heat could be generated by mechanical work. This lead heat to be a form of energy transfer.

1-4C

How do rating problems in heat transfer differ from the sizing problems?

Rating problems, the size and geometry of a system are known, and the goal is to determine the rate of heat transfer under operating conditions.

Sizing problems, the desired heat transfer rate is specified, and the goal is to determine the required size/dimension of the system to meet the requirements.

1-5C

What is the difference between the analytical and experimental approach to heat transfer? Discuss the advantages and disadvantages of each approach.

Analytical approach involves solving equations using mathematical analysis to obtain solutions.

The advantage is that it provides a general solution given various parameters. The disadvantage is that it is often limited to simple geometric shapes and boundary conditions.

Experimental approach involves measurements and observations from experiments. The advantage is that it can be applied to complex shapes and real conditions. The disadvantage is that it is case-specific and can be costly/time consuming as well as produce measurement errors.

1-6C

How do the differential equations in the study of a physical problem arise?

Differential equations arise from applying conservation principles, like conservation of energy, to a differential control volume.

1-7C

What is the importance of modeling in engineering? How are the mathematical models for engineering processes prepared?

Modeling allows engineers to analyze, predict, and understand the behavior of real systems.

Mathematical models are prepared by making simplifying assumptions that describe the system.

1-8C

When modeling an engineering process, how is the right choice made between a simple but crude and a complex but accurate model? Is the complex model necessarily a better choice since it is more accurate?

The choice is made by considering the purpose of the analysis, accuracy, time, and resources. A simple model may be sufficient if it captures the effects/results with acceptable accuracy.

A complex model is not necessarily a better choice since it can require more time, resources, and effort to get similar results to a simpler model.

1-9C

On a hot summer day, a student turns his fan on when he leaves his room in the morning. When he returns in the evening, will his room be warmer or cooler than the neighboring rooms? Why? Assume all the doors and windows are kept closed.

The room will be warmer than the neighboring rooms. The fan adds electrical energy to the room which is converted into thermal energy resulting in an increase in the room's internal energy and temperature.

1-10C

Consider two identical rooms, one with a refrigerator in it and the other without one. If all the doors and windows are closed, will the room that contains the refrigerator be cooler or warmer than the other room? Why?

The room will be warmer than the neighboring rooms. The refrigerator removes heat from its interior and rejects the heat to the room resulting in additional energy which increase the total thermal energy of the room, raising the temperature.

1-11C

An ideal gas is heated from 50°C to 80°C (a) at constant volume and (b) at constant pressure. For which case do you think the energy required will be greater? Why?

The energy required will be greater for constant pressure. When gas is heated at constant pressure the gas expands the system and increases internal energy. At constant volume there is no work supplied to the system, and it all goes into internal energy.

1-12C

How are heat, internal energy, and thermal energy related to each other?

Internal energy is a property of a system that represents the energy stored within the system.

Thermal energy is the portion of internal energy associated with the motion of molecules and is related to temperature.

Heat is not a property of a system, it is energy in transit across a system because of a temperature difference. Heat transfer changes a system's internal energy and therefore its thermal energy component.

1-13C

What is heat flux? How is it related to the heat transfer rate?

Heat flux is the rate of heat transfer per unit surface area normal to the direction of heat flow.

The heat transfer rate is related to heat flux by multiplying the heat flux by the surface area.

1-14C

What are the mechanisms of energy transfer to a closed system? How is heat transfer distinguished from the other forms of energy transfer?

The mechanisms of energy transfer to a closed system is by heat transfer and work.

Heat transfer only occurs because of a temperature difference between the system and its surroundings. Work is energy transfer associated with forces crossing a systems boundaries.

Heat Transfer Mechanisms

1-33C

Define thermal conductivity and explain its significance in heat transfer.

Thermal conductivity is a material property that indicates a material's ability to conduct heat.

Thermal conductivity is significant because it affects the rate of heat conduction.

1-34C

Judging from its unit $W/m \cdot K$, can we define thermal conductivity of a material as the rate of heat transfer through the material per unit thickness per unit temperature difference? Explain.

No. Thermal conductivity cannot be defined by itself as the rate of heat transfer since it is a material property. The rate of heat transfer is determined by Fourier's Law.

1-35C

Which is a better heat conductor, diamond or silver?

Diamond is a better heat conductor than silver due to its high thermal conductivity.

1-36C

How do the thermal conductivity of gases and liquids vary with temperature?

Thermal conductivity of gases tends to increase with temperature.

Thermal conductivity of liquids tends to decrease with temperature.

1-37C

Why is the thermal conductivity of superinsulation orders of magnitude lower than the thermal conductivity of ordinary insulation?

The thermal conductivity of superinsulation orders of magnitude is lower due to the minimization of heat transfer via conduction and radiation.

1-38C

Why do we characterize the heat conduction ability of insulators in terms of their apparent thermal conductivity instead of the ordinary thermal conductivity?

Insulators are characterized by their apparent thermal conductivity because it combines conduction, convection, and radiation. Ordinary thermal conductivity represents only conduction.

1-39C

Consider an alloy of two metals whose thermal conductivities are k_1 and k_2 . Will the thermal conductivity of the alloy be less than k_1 , greater than k_2 , or between k_1 and k_2 ?

The thermal conductivity will be between k_1 and k_2 since this represents both materials thermal conductivity.

1-40C

What are the mechanisms of heat transfer? How are they distinguished from each other?

Conduction is heat transfer due to molecular interactions.

Convection is heat transfer between a surface and moving liquid.

Radiation is heat transfer through electromagnetic waves

1-41C

Write down the expressions for the physical laws that govern each mode of heat transfer, and identify the variables involved in each relation.

Conduction governed by Fourier's Law, which relates heat transfer rate to the temperature gradient, thermal conductivity, and area of heat flow.

Convection governed by Newton's Law of Cooling, which relates heat transfer rate to convection heat transfer coefficient, surface area, and temperature difference between surface and liquid.

Radiation governed by Stefan-Boltzmann Law, which relates heat transfer rate to emissivity, surface area, and fourth power of the absolute temperature.

1-42C

How does heat conduction differ from convection?

Heat conduction is the transfer of energy due to molecular interactions.

Convection involves heat transfer between a surface and moving fluid resulting in both molecular conduction and bulk fluid motion.

1-43C

Does any of the energy of the sun reach the earth by conduction or convection?

No. The energy from the sun reaches earth only by radiation, since the space between the sun and the earth is a vacuum which prevents heat transfer by conduction or convection.

1-44C

How does forced convection differ from natural convection?

Forced convection occurs when the fluid motion is caused by external means, such as a pump or fan.

Natural convection occurs when fluid motion is caused by buoyancy forces due to temperature variations causing density differences.

1-45C

What is the physical mechanism of heat conduction in a solid, a liquid, and a gas?

Solid heat conduction occurs due to lattice vibrations of atoms.

Liquid heat conduction occurs due to molecular interactions and diffusions.

Gas heat conduction occurs due to random molecular motion and collision

1-46C

Consider heat transfer through a windowless wall of a house on a winter day. Discuss the parameters that affect the rate of heat conduction through the wall.

The rate of heat conduction through the wall is affected by thermal conductivity of the wall material, thickness, surface area, and temperature difference.

1-47C

Consider heat loss through the two walls of a house on a winter night. The walls are identical, except that one of them has a tightly fit glass window. Through which wall will the house lose more heat? Explain.

The house will lose more heat through the wall with the glass window, since glass has a higher thermal conductivity. There will be greater heat loss at the window than the insulated wall.

1-48C

Consider two houses that are identical, except that the walls are built using bricks in one house, and wood in the other. If the walls of the brick house are twice as thick, which house do you think will be more energy efficient?

The house with wood walls will be more energy efficient, since the brick walls have a higher thermal conductivity causing greater heat loss.

1-49C

Consider two walls of a house that are identical except that one is made of 10-cm-thick wood, while the other is made of 25-cm-thick brick. Through which wall will the house lose more heat in winter?

The house will lose more heat through the brick wall since it has higher thermal conductivity causing greater heat loss. The low thermal conductivity of wood provides a greater resistance to heat flow.

1-50C

Define emissivity and absorptivity. What is Kirchhoff's law of radiation?

Emissivity is a measure of a surface ability to emit thermal radiation compared to a blackbody.

Absorptivity is the fraction of the incident radiation on a surface that is absorbed by the surface.

Kirchhoff's Law of Radiation states, for a surface in thermal equilibrium, the emissivity equals the absorptivity at the same temperature and wavelength.

1-51C

What is a blackbody? How do real bodies differ from blackbodies?

Blackbody is an idealized surface that absorbs all incident radiation and emits the maximum possible radiation.

Real bodies do not absorb or emit all incident radiation. Their behavior is characterized by an emissivity less than one.

Simultaneous Heat Transfer Mechanisms

1-90C

Can all three modes of heat transfer occur simultaneously (in parallel) in a medium?

Yes. Conduction, convection, & radiation can all occur simultaneously.

1-91C

Can a medium involve (a) conduction and convection, (b) conduction and radiation, or (c) convection and radiation simultaneously? Give examples for the "yes" answers.

Conduction and convection can occur together, such as heat transfer from a solid surface to a moving liquid.

Conduction and radiation can occur together, such as heat transfer through a solid wall where heat is conducted and radiated from the surface.

Convection and radiation can occur together, such as heat transfer from a hot surface to the surrounding air.

1-92C

The deep human body temperature of a healthy person remains constant at 37 ° C while the temperature and the humidity of the environment change with time. Discuss the heat transfer mechanisms between the human body and the environment both in summer and winter, and explain how a person can keep cooler in summer and warmer in winter.

The heat transfer mechanisms that the human body uses is conduction, convection, radiation, and evaporation.

Summer, the body evaporates sweat which removes heat from the skin, as well as radiation from the surrounding surface increase the temperature. A person can prevent this by wearing lighter clothes to lower the effect of radiation and to allow heat and moisture to escape.

Winter, the body loses heat due to conduction and convection to the colder air and by radiation to colder surrounding surfaces. A person can prevent this by wearing insulating clothes to reduce conduction and convection, as well as limiting exposed surfaces which will reduce radiative heat loss.

1-93C

We often turn the fan on in summer to help us cool. Explain how a fan makes us feel cooler in the summer. Also explain why some people use ceiling fans also in winter.

Summer, a fan makes us feel cooler by increasing air motion. This evaporates sweat from the skin and increases convective heat transfer which increases the rate of heat loss.

Winter, ceiling fan circulate warm air that rises towards the ceiling. By improving air mixing the fan helps uniformly heat the room while not changing the room temperature.